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Prediction Of Maternal Health Risk Factors Using Machine Learning Algorithms

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Abstract

Ensuring the well-being of both mother and child during pregnancy requires diligent monitoring of maternal health. This study explores the application of machine learning (ML) models to predict maternal health risks, using a dataset of 1014 instances with six health metrics. The risk categories- low, medium and high- were predicted using several ML techniques, including Ensemble (Bagged Trees), k-NN, and Neural Networks. Each model was evaluated based on its accuracy, confusion matrices, and other key metrics. The Ensemble Bagged Trees method achieved the highest accuracy of 84.12%, outperforming other models. The findings in this paper demonstrate the potential of these ML techniques to provide timely and accurate risk assessments for maternal health risk in pregnant women.

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Keywords: Machine Learning; Classification; Maternal Health; Accuracy; Bagged Trees

1. Introduction

Maternal health, encompassing the care from pregnancy through the postpartum period, is crucial for ensuring the well-being of women and their infants. Throughout this continuum, various risks can profoundly impact maternal and foetal health. Obstetric emergencies like haemorrhage, hypertensive disorders, sepsis, abortion, obstructed labour, ectopic pregnancy, and embolism pose immediate threats to maternal health [1]. Notably, postpartum haemorrhage is the main cause of maternal death globally [2]. About 24% of haemorrhage-related deaths occur during pregnancy [3]. Improved emergency obstetric care at birth is crucial intervention. To achieve substantial reductions in maternal mortality, it is essential to strengthen healthcare systems to provide these services [4,5]. Despite global efforts,

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maternal mortality remains a significant concern, particularly in areas with limited healthcare access, underscoring persistent disparities.

Machine learning (ML) holds immense promise in revolutionizing healthcare by leveraging big data to improve diagnostic accuracy and treatment outcome [6]. ML algorithms examine diverse maternal health indicators like blood pressure, glucose levels, maternal age, medical history, and socio-economic, uncovering complex correlations often missed by traditional methods. A key application of ML in maternal health is risk stratification based on pregnant women's health profiles, enabling healthcare providers to prioritize interventions for high-risk individuals promptly and effectively. Moreover, ML algorithms can predict specific complications like postpartum haemorrhage early in pregnancy, and can also predict risk of postpartum depression [7,8]. ML models continuously refine their predictions with new data, improving the accuracy of risk assessments and providing timely insights for informed clinical decisions, thereby contributing to better maternal health outcomes globally. Integrating ML with health administrative data enhances risk prediction and management, potentially leading to improved maternal health outcomes [9]. Through a detailed comparison of several ML models, this work identifies the most effective algorithms for predicting maternal health risks, offering a valuable resource for improving clinical decision-making in maternal healthcare. This study contains a comprehensive comparison of various ML models in predicting maternal health risks using a diverse dataset.

In the remainder of this paper, Section 2 details the dataset and preprocessing methods, followed by Section 3, where we describe the methodology used for model development. Section 4 presents the experimental results and discussion of the performance of models. Finally, Section 5 concludes the paper with a summary of key findings and suggestions for future research.

2. Input data

The dataset, collected through an IoT-based risk monitoring system from rural areas of Bangladesh, provides a critical resource for predicting maternal health risks [10]. It comprises 1014 instances and includes six significant health metrics which are all integer values. These metrics are utilized to predict risk intensity (RiskLevel) during pregnancy, categorized as low, medium, and high risk. Each feature offers valuable insights into maternal health. Age, measured in years, is vital for understanding age-related risks such as gestational diabetes and hypertension, which can significantly impact pregnancy outcomes. The dataset includes individuals aged between 10 to 70 years, with a predominant frequency of younger mothers, particularly between 20 and 30 years. SystolicBP and DiastolicBP, represent upper and lower blood pressure values in mmHg respectively. Monitoring BP is essential for detecting hypertensive disorders such as gestational hypertension and preeclampsia, which are associated with adverse outcomes for both the mother and the foetus [11]. In the dataset, SystolicBP ranges from 70 to 160 mmHg, while DiastolicBP ranges from 49 to 100 mmHg. During pregnancy, SystolicBP typically ranges from approximately 104 to 116 mmHg, and DiastolicBP generally ranges from approximately 61 to 65 mmHg [11]. Blood Sugar (BS), recorded in mmol/L, is also known as glucose level. It is crucial for assessing glucose levels and providing insights into the risk of diabetes, which can pose significant risks to both mother and foetus if unmanaged. BS levels in the dataset range from 6 to 19 mmol/L. Body Temperature (BodyTemp), measured in Fahrenheit, and Heart Rate, noted in beats per minute (bpm), serve as vital signs reflecting the mother's physiological state and can indicate underlying health issues affecting pregnancy. The dataset includes BodyTemp values ranging from 98 to 103˚F and Heart Rate ranging from 7 to 90 bpm. Heart Rate generally ranges from approximately 74 to 79 bpm for pregnant women depending on the gestation age [11]. The histogram plots for the predictor variables are shown in Figure 1.

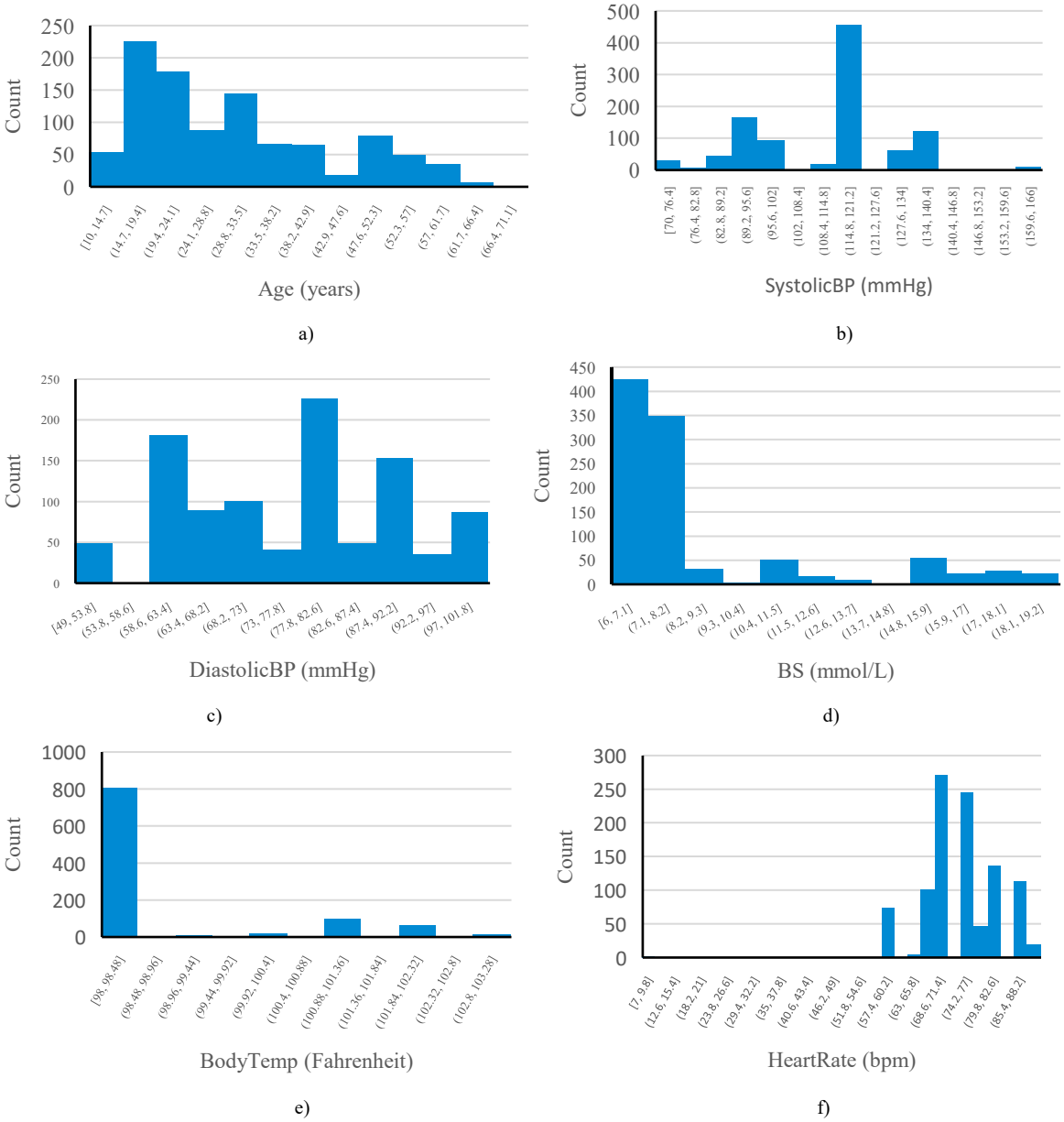


Fig. 1. Histogram Plots for various predictor variables: a) Age, b) SystolicBP, c) DiastolicBP, d) BS, e) BodyTemp, f) HeartRate

3. Machine learning classifiers

This section provides a concise summary of the various ML Techniques used in the present study.

3.1. Ensemble

Ensemble Methods in ML combine predictions from multiple models to enhance overall performance and robustness, typically outperforming individual models alone. They follow the principle “Many are smarter than few”. Ensemble methods aggregate predictions from multiple models, leveraging their collective insights to get more

accurate predictions than those of individual models. By enhancing the reliability and generalizability of ML models across diverse datasets and conditions.

Ensemble methods include Voting or averaging, Bagging, Random Forest, Boosting, and Stacking. Boosting is a technique where models are built sequentially, each one correcting error made by its predecessors. It emphasizes training instances that were previously misclassified by adjusting their weights so that subsequent models focus more on these difficult cases [12]. This sequential approach creates a robust learner from a series of weak learners by minimizing errors and improving model accuracy. Stacking combines multiple models (known as base models) to enhance predictive performance. It involves training a new model to efficiently blend the predictions made by these base models.

Bagging (Bootstrap Aggregation) is a technique designed to enhance the stability and accuracy of ML algorithms, particularly those susceptible to high variance, like decision trees. The process begins by generating multiple training subsets, known as bootstrap samples, from the original dataset through random sampling with replacement [12]. Separate models are then trained on each of these samples, capturing different patterns. Once all models are trained, their predictions are aggregated by averaging (regression) or majority voting (classification) to produce a final output, as shown in Figure 2. Bagging offers several advantages, including variance reduction, improved accuracy, and increased robustness against noise and outliers. It creates more stable and reliable predictive model that often outperforms any single model trained on the entire dataset.

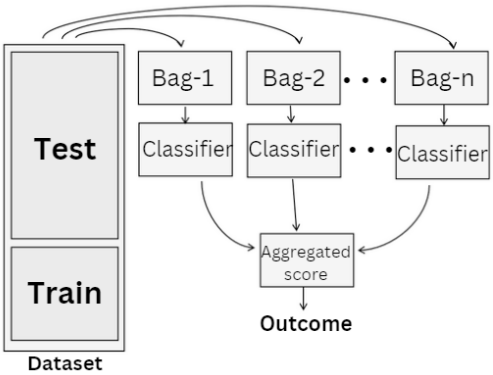


Fig. 2. Work flow of Bagged Trees

3.2. k-Nearest Neighbors (k-NN)

k-Nearest Neighbors (k-NN) algorithm is utilized for regression as well as classification. It identifies the "k" nearest neighboring dataset points for the given dataset point. The nearest data point distance is generally calculated using the Euclidean distance formula. It forecasts the class of a given data point by determining the most common class among the “k” nearest neighbors.

The dataset's characteristics help in the selection of the value of “k”. Rather than using a single global k, fine k-NN adjusts k according to local data traits or different sub-regions of the dataset. Weighted k-NN is another variant which assigns the weights to neighbors based on their distance from the query point.

3.3. Naïve Bayes

Naïve Bayes classifiers are algorithms that apply Bayes’ theorem to address classification problems. They are based on the principle of conditional probability and operate under the assumption of feature independence, hence the term ‘naïve’. Naïve Bayes classifiers frequently achieve strong performance in practice, even though they rely on the straightforward assumption.

The ‘naïve’ assumption simplified the calculation of the likelihood *P(C|X)*:

$$P(X|C) = P(X_1|C).P(X_2|C) \dots P(X_n|C) \quad (1)$$

3.4. Neural Networks

Neural networks, designed after the architecture of the human brain, are essential in ML. These networks consist of a series of algorithms intended to uncover underlying relationships within a dataset, emulating the function of the human brain [13].

The networks consist of interconnected layers of neurons, comprising input, hidden, and output layers. Data flows into the neural network via the input layer, traverses through hidden layers where neurons process and identify patterns, and exists through the output layer. Each neuron performs computations based on received data, applies weights, and uses an activation function (like ReLU, Sigmoid, or Tanh).

Neural networks are designated in various architectures tailored to accommodate different types of data. Wide and Medium Neural Networks are a few types of architectures. Wide Neural Networks have larger layers that are hidden, enabling them to capture complex patterns directly. They are particularly effective in scenarios requiring the modeling of broad and intricate interactions within the data. Medium neural networks strike a balance between wide and deep architectures, typically having moderate-sized layers. They are versatile and capable of addressing a diverse array of tasks while providing a good balance between complexity and computational efficiency.

4. Results

This section provides the performance results of different ML models assessed for predicting pregnancy health risks across low, medium, and high-risk categories. It includes model accuracy, confusion matrices, evaluation metrics, and details on hyperparameters.

4.1. Model performance

Ensemble (Bagged Trees) achieved the highest accuracy at 84.12%, demonstrating its superior performance in classifying maternal health risks. Fine KNN (83.73%) and Weighted KNN (82.94%) also showed high accuracy, making them viable alternatives. Neural Network models, such as Wide Neural Network (82.64%) and Medium Neural Networks (79.68%), performed robustly, reflecting their capacity to handle complex maternal health data.

Figure 3 provides a visual comparison of validation accuracy among the tested models. It clearly shows that the Ensemble (Bagged Trees) outperforms other models, with Fine KNN and Weighted KNN closely following. This figure highlights the effectiveness of ensemble techniques and KNN variations over simpler models such as SVM and Logistic Regression.

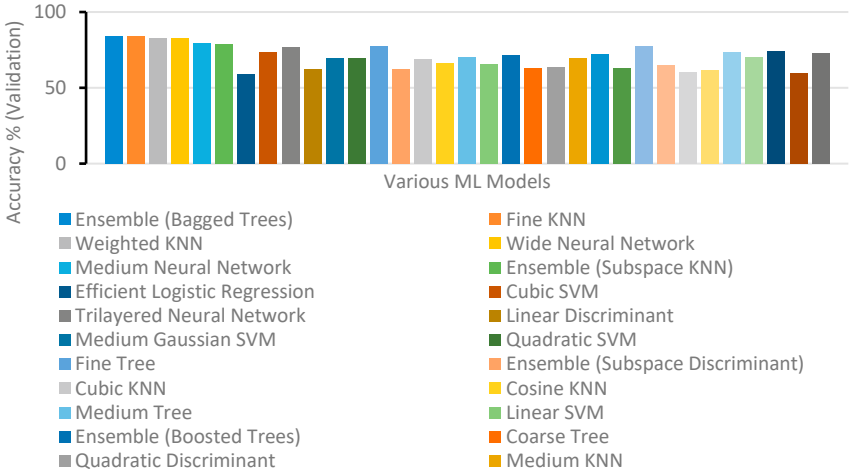


Fig. 3. Comparison of Validation accuracy across various ML models

Figure 4 illustrates the Receiver Operating Characteristic (ROC) curve, which evaluates the performance of the 6 best-performing ML models. A perfect classifier would produce a curve that is close to the top left corner of the plot, showing a high true positive rate and a low positive rate. As we can see, Ensemble (Bagged Trees) and Fine KNN models have the highest accuracy as their ROC curves are closest to the top left.

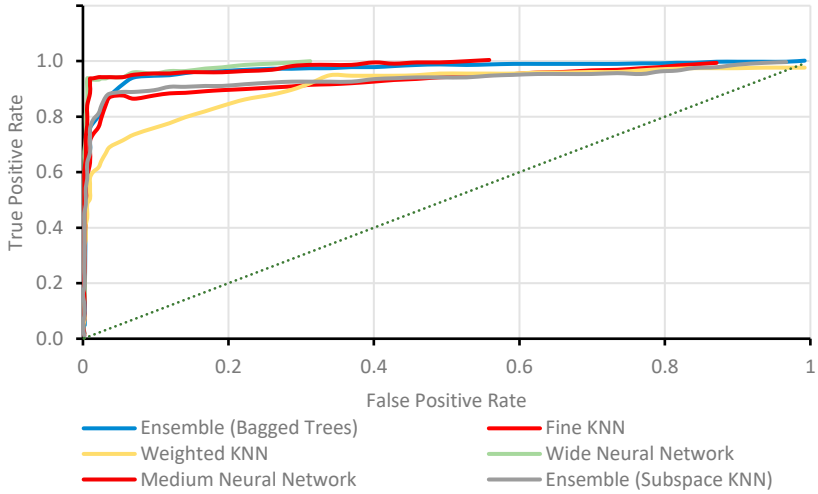
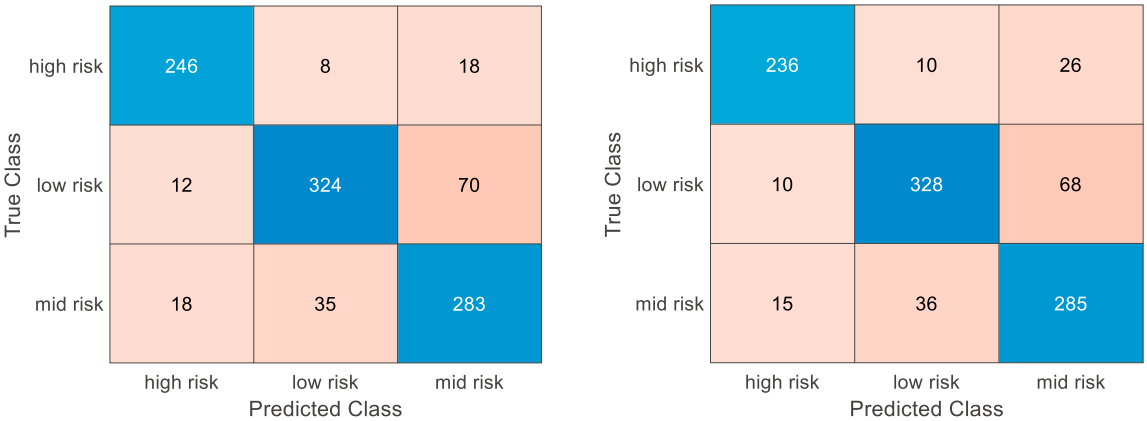


Fig. 4. ROC curves of the 6 best-performing ML models

4.2. Confusion matrices

Confusion matrices for the 6 best-performing models (Ensemble Bagged Trees, Fine KNN, weighted KNN, Wide Neural Network, Medium Neural Network, Ensemble Subspace KNN) are illustrated in Figure 5. Analysis reveals that the Ensemble (Bagged Trees) model has the highest True Positive (TP) and True Negative (TN) rates, and lowest False Positive (FP) and False Negative (FN) rates, demonstrating its accuracy and reliability in classifying maternal health risks.



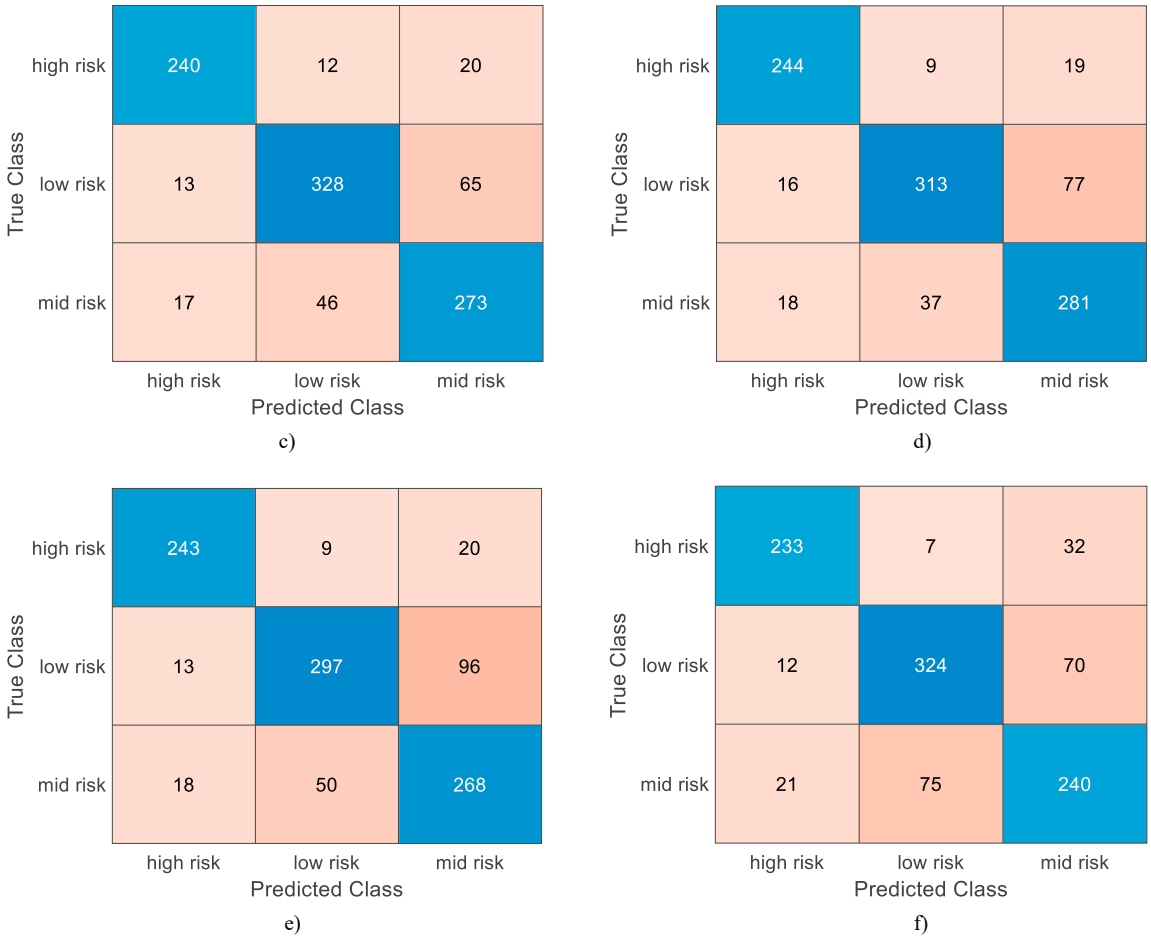


Fig. 5. Confusion matrices for the 6 best-performing ML models with the highest accuracy. a) Ensemble (Bagged Trees), b) Fine KNN, c) Weighted KNN, d) Wide Neural Network e) Medium Neural Network, f) Ensemble (Subspace KNN)

4.3. Evaluation metrics

Table 1 displays various evaluation metrics, illustrating how they differ among the 6 best-performing ML models.

The accuracy metric measures the proportion of correct predictions out of the total number of instances evaluated [14].

$$Accuracy = \frac{\sum Diagonal\ Elements}{\sum All\ Elements} \tag{2}$$

Precision is the ratio of true positive predictions to the total number of positive predictions [14].

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

where *TP* is True Positive, and *FP* is False Positive.

Recall is the proportion of true positives out of all actual positives [14].

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

where *TP* is True Positive and *FN* is False Negative.

F1-score is the harmonic mean of precision and recall, offering a balance between the two metrics [14].

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

Table 1. Performance metrics of the 6 best-performing ML models

Evaluation metrics \ Models		Ensemble (Bagged Trees)	Fine KNN	Weighted KNN	Wide Neural Network	Medium Neural Network	Ensemble (Subspace KNN)
Accuracy % (Validation)		84.12	83.73	82.94	82.64	79.68	78.6
Precision (in %)	High	89.1	90.4	88.9	87.8	88.7	87.6
	Low	88.3	87.7	85	87.2	83.5	79.8
	Mid	76.3	75.2	76.3	74.5	69.8	70.2
Recall (in %)	High	90.4	86.8	88.2	89.7	89.4	85.7
	Low	79.8	80.8	80.8	77.1	73.2	79.8
	Mid	84.2	84.8	81.2	83.7	79.8	71.4
F1-score (in %)	High	89.7	88.6	88.5	88.7	89	86.7
	Low	83.8	84.1	82.8	81.8	77.9	79.8
	Mid	80	79.7	78.7	78.8	74.4	70.8

While accuracy is a key metric, this paper places significant emphasis on recall and precision, particularly for high-risk classifications. In healthcare settings, especially for maternal health, minimizing false negatives is crucial, as misclassifying high-risk individuals could lead to delayed or insufficient interventions. The Bagged Trees and Neural Networks models excel in this regard, offering a higher recall rate compared to traditional models [15].

Figure 6 is a bar chart that presents a comparative analysis of the 6 best-performing ML models, evaluating their performance in terms of F1-score, Recall, and Precision.

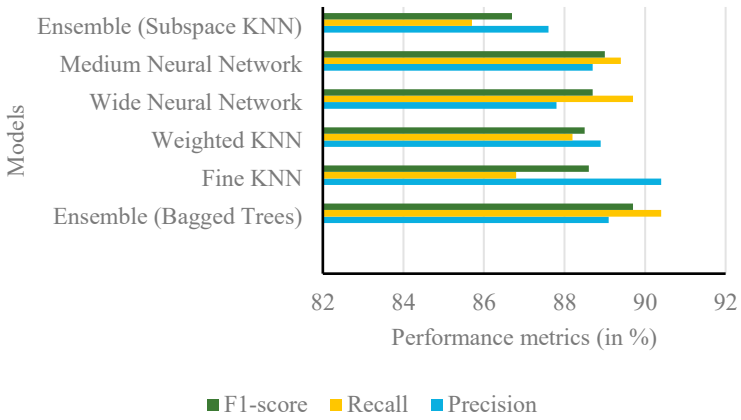


Fig. 6. Comparison of performance metrics

4.4. Hyperparameters

Hyperparameter tuning significantly influenced model performance. As shown in Table 2, Ensemble (Bagged Trees) utilized a high number of learners and splits, enhancing its classification capabilities. KNN models varied in the number of learners as splits, enhancing its classification capabilities. KNN models varied in the number of

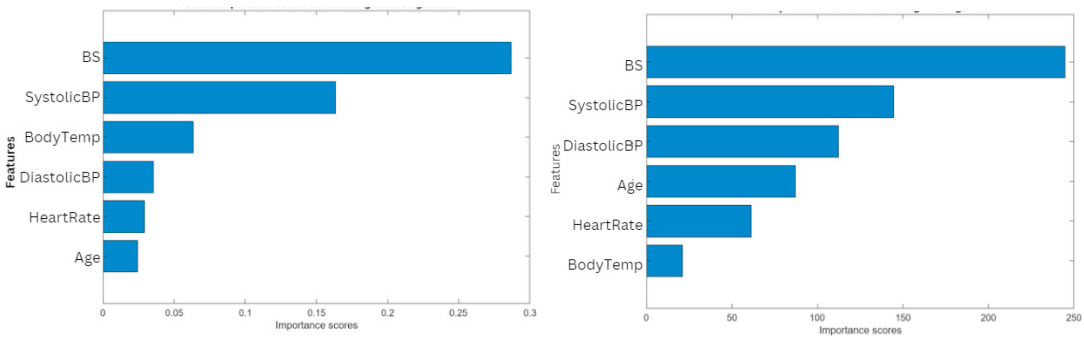
neighbours and distance metrics, optimizing their performance based on the dataset's characteristics. Neural Network models were fine-tuned with different layer sizes and activation functions to balance complexity and generalization.

Table 2. List of hyperparameters used to optimize the 6 best-performing ML models

ML models	Parameters Used
Ensemble (Bagged Trees)	Learner type: Decision tree, Maximum number of splits: 1013, Number of learners: 30
Fine KNN	Number of neighbours: 1, Distance metric: Euclidean, Distance weight: Equal, Standardize data: Yes
Weighted KNN	Number of neighbours: 10, Distance metric: Euclidean, Distance weight: Squared inverse, Standardize data: Yes
Wide Neural Network	Number of fully connected layers: 1, First layer size: 100, Activation: ReLU. Iteration limit: 1000, Regularization strength (Lambda): 0, Standardize data: Yes
Medium Neural Network	Number of fully connected layers: 1, First layer size: 25, Activation: ReLU, Iteration limit: 1000, Regularization strength (Lambda): 0, Standardization data: Yes
Ensemble (Subspace KNN)	Learner type: Nearest neighbours, Number of learners: 30, Subspace dimension: 3

4.5 Feature Selection

The figure 7 represents feature importance scores for various maternal health indicators using MRMR and Chi-square algorithms. BS (Blood Sugar) consistently ranks highest in both methods, indicating its strong relevance to predict maternal health risks. SystolicBP is also crucial according to both methods, emphasizing the importance of blood pressure measurements.



a) b)

Fig. 7. Feature important scores sorted using a) MRMR algorithm b) Chi2 algorithm

5. Conclusion

This study investigated the efficiency of ML algorithms in predicting maternal health risks, highlighting their potential to improve maternal care by accurately classifying risk levels. By analysing a dataset of 1014 instances with six health metrics, several ML models such as Naïve Bayes, k-Nearest Neighbours (k-NN), Ensemble Models, Neural Networks, etc. were evaluated. This approach provided valuable insights into how diverse ML techniques can handle different health indicators to identify and stratify risks efficiently.

- The Bagged Trees ensemble method demonstrated the best performance, achieving an accuracy of 84.12%, thus proving effective in classifying maternal health risks and handling diverse data inputs.
- Neural Network models, particularly Wide and Medium architectures, also showed robust performance, underscoring their capability to handle complex interactions within maternal health data efficiently.

- Hyperparameter tuning is essential for enhancing model performance, as evidenced by the optimized parameters leading to significant accuracy improvements.

Machine Learning models, particularly ensemble methods like Bagged Trees, offer promising tools for early and accurate prediction of maternal health risks. These findings emphasize the potential of ML-driven approaches to improve maternal health monitoring especially in resource-limited settings where timely risk assessment can be life-saving. The predictive capabilities of these models can empower healthcare systems to focus resources on high-risk cases, ultimately contributing to reduced maternal mobility and improved maternal health globally. As healthcare systems increasingly rely on data-driven solution, ML models like the ones evaluated in this study can become integral to preventive healthcare strategies.

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